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X 'For You' Algorithm

Deep-Dive Analysis Report

A comprehensive reverse-engineering of the open-source X recommendation system. Covers Phoenix (Grok-based transformer ranking), Thunder (in-network store), Home Mixer (orchestration), and Grox (content understanding).

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Source: x-algorithm repository (May 2026 release)

Components: home-mixer/ (Rust), phoenix/ (Python/JAX), thunder/ (Rust), grox/ (Python)

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Pipeline Stages

10
Pre-Score Filters

15+
Engagement Signals

5
Negative Signals

1. How the Algorithm Works

The For You feed is assembled through a multi-stage pipeline. Every tweet you see has survived candidate sourcing, hydration, filtering, ML-based scoring by a Grok-based transformer (Phoenix), diversity adjustment, and final selection.

Stage	What Happens	Key Detail
1. Query Hydration	Fetch user context	Engagement history, follows, mutes, blocks, topics, IP
2. Candidate Sourcing	Retrieve candidates	Thunder (in-network) + Phoenix Retrieval (out-of-network)
3. Hydration	Enrich metadata	Author info, media, video duration, engagement counts, mutual follows
4. Pre-Score Filtering	Remove ineligible	Duplicates, old posts, self-posts, blocked/muted, seen, muted keywords
5. Phoenix Scoring	ML predictions	Grok transformer predicts 15+ action probabilities per tweet
6. Weighted Scoring	Combine predictions	Final Score = SUM(weight x P(action)) across all action types
7. Author Diversity	Decay repeats	Exponential decay multiplier for repeated authors
8. OON Adjustment	Balance networks	Out-of-network tweets get a scaling factor
9. Selection	Pick top K	Sort by final score, select top candidates
10. Post-Selection	Safety check	Visibility filtering, conversation dedup

The Core Ranking Formula

Final Score = SUM(weight_i x P(action_i)) -- Phoenix predicts the probability of each engagement type (like, reply, repost, share, video view, dwell, follow, block, mute, report, etc.) and these are multiplied by configurable weights. Positive actions boost; negative actions suppress.

2. Positive Signals (What Boosts Your Tweet)

Each signal below has its own weight in the scoring formula. More signal types triggered = higher combined score.

Signal	Code Reference	Impact
P(favorite)	favorite_score x FAVORITE_WEIGHT	Very High -- likes are primary
P(reply)	reply_score x REPLY_WEIGHT	Very High -- conversation signal
P(repost)	retweet_score x RETWEET_WEIGHT	Very High -- share-worthy content
P(quote)	quote_score x QUOTE_WEIGHT	High -- thought-provoking
P(share)	share_score x SHARE_WEIGHT	High -- general sharing
P(share_via_DM)	share_via_dm_score x SHARE_VIA_DM_WEIGHT	High -- personal value
P(share_copy_link)	share_via_copy_link_score x weight	High -- off-platform sharing
P(click)	click_score x CLICK_WEIGHT	Medium -- expanding/links
P(profile_click)	profile_click_score x weight	Medium -- profile visits
P(photo_expand)	photo_expand_score x weight	Medium -- image engagement
P(video_view)	vqv_score x VQV_WEIGHT	Medium-High -- video only, duration gated
P(dwell)	dwell_score x DWELL_WEIGHT	Medium -- stopped scrolling
P(dwell_time)	dwell_time x CONT_DWELL_TIME_WEIGHT	Medium -- reading duration
P(follow_author)	follow_author_score x weight	High -- conversion signal

3. Negative Signals (What Kills Your Tweet)

Signal	Code Reference	Impact
P(not_interested)	not_interested_score x NOT_INTERESTED_WEIGHT	Strong Negative
P(block_author)	block_author_score x BLOCK_AUTHOR_WEIGHT	Very Strong Negative
P(mute_author)	mute_author_score x MUTE_AUTHOR_WEIGHT	Strong Negative
P(report)	report_score x REPORT_WEIGHT	Very Strong Negative
P(not_dwelled)	not_dwelled_score x NOT_DWELLED_WEIGHT	Moderate Negative

When a tweet's combined score goes negative, it is rescaled: $(\text{score} + \text{NEGATIVE_WEIGHTS_SUM}) / \text{WEIGHTS_SUM} \times \text{NEGATIVE_SCORES_OFFSET}$. The model predicts these for EVERY candidate -- if your past content has a pattern of triggering blocks, the model generalizes this to your future content.

4. Content Types the Algorithm Favors

Video: Exclusive VQV Scoring

Videos have a dedicated VQV (Video Quality View) signal with its own weight. Only videos longer than MIN_VIDEO_DURATION_MS qualify -- the code explicitly sets vqv_weight = 0.0 for short clips. Thunder maintains a separate video_posts_by_user index. Quoted tweets with video also get special treatment via quoted_vqv_weight.

Images: Dedicated Expansion Signal

Photo expansion (photo_expand_score) is a separate prediction head. The model specifically learns whether a user will click to expand an image. Posts with compelling images that users expand get boosted. HasMediaHydrator detects media presence as a boolean feature for the ML model.

Dwell Time: Reading Duration is Measured

Three dwell signals exist: binary dwell (stopped scrolling), continuous dwell_time (how long), and click_dwell_time (time after clicking in). Write longer, substantive posts that reward careful reading.

5. Grox Content Safety Pipeline

Before tweets reach ranking, the Grox system uses Grok VLM classifiers to evaluate content quality and safety.

Classifier	What It Does	Impact
BangerInitialScreen	Assigns quality_score (0-1), slop_score, tags, tab_order	Below 0.4 quality = flagged
SpamClassifier	Targets low-follower accounts for spam detection	Spam = removed from feed
SafetyPTOS Category	Checks 7 safety policy categories	Violations dropped by VFFilter
SafetyPTOS Policy	Per-policy analysis with reasoning fallback	ViolentMedia, AdultContent, Spam, Hate, etc.
PostSafetyDeluxe	Extended safety screening	Produces tweet_bool_metadata
ReplyScorer	Scores reply quality 0-3 scale	Determines reply visibility

6. How to Maximize Your Reach on X

These strategies are derived directly from how the scoring algorithm works in production code.

Strategy 1: Maximize Multi-Signal Engagement

A tweet with 50 likes + 20 replies + 10 reposts scores higher than 100 likes alone. The formula is a SUM across all weighted predictions. More signal types = higher combined score.

Strategy 2: Drive DM + Copy Link Shares

Three separate share signals exist. Content people send to friends or copy-paste elsewhere gets a triple share boost. Create 'you need to see this' content.

Strategy 3: Optimize for Dwell Time

Binary dwell, continuous dwell_time, and click_dwell_time all contribute independently. Write substantive posts. Include details that reward careful reading.

Strategy 4: Use Video Above Minimum Duration

VQV weight is ONLY applied when duration exceeds MIN_VIDEO_DURATION_MS. Below = 0.0. This is a bonus signal text and image posts cannot receive.

Strategy 5: Trigger Follow Actions

P(follow_author) is directly in the scoring formula. Establish clear expertise. The algorithm rewards accounts that convert impressions to follows.

Strategy 6: Post Consistently

Phoenix processes the last 128 engagement actions. Regular posting builds patterns the transformer relies on. Irregular posting breaks the signal.

Strategy 7: Leverage Mutual Follow Networks

Jaccard similarity between your followers and a viewer's followers (256 MinHash values) affects out-of-network relevance. Engage with communities in your niche.

Strategy 8: Nail the First Hour

Post age buckets have 60-minute granularity, max 4800 minutes. Thunder retention = 2 days. Strong engagement in the first bucket signals relevance and compounds.

7. The Don'ts -- What Kills Your Reach

Don't	Why (From Code)	Severity
Trigger blocks/mutes/reports	P(block), P(mute), P(report) have strong NEGATIVE weights. Model generalizes from past patterns.	CRITICAL
Make people scroll past	P(not_dwelted) is a negative signal. Consistently skipped content trains the model to demote you.	HIGH
Flood the feed	AuthorDiversityScorer: exponential decay. multiplier = $(1 - \text{floor}) * \text{decay}^{\text{position}} + \text{floor}$.	HIGH
Post spam / low-quality	Grox SpamClassifier + BangerInitialScreen. quality < 0.4 = flagged. Lslop_score is real.	CRITICAL
Use muted keywords	MutedKeywordFilter tokenizes text and matches. Tweet invisible to anyone who muted a keyword.	MEDIUM
Violate safety policies	VFFilter removes flagged tweets. 7 policy categories checked by SafetyPTOS.	CRITICAL
Post short throwaway videos	Videos below MIN_VIDEO_DURATION_MS: vqv_weight + MED. Entire VQV scoring signal lost.	MEDIUM
Ignore early engagement	Phoenix 128-action history window. No early engagement = not in histories = not in future ranking.	HIGH
Trigger 'Not Interested'	Dedicated negative signal. Pattern of 'Not Interested' creates a downward spiral.	HIGH

8. Key Technical Insights

Candidate Isolation

In the Phoenix transformer, candidates cannot attend to each other -- only to user context. Your tweet is scored independently. Scores are purely a function of (your content x this specific user's history).

Hash-Based Identity

Users and posts are represented through multiple hash embeddings (2 user, 2 item, 2 author hashes default). The model doesn't see your username -- it sees embedding patterns learned from engagement.

New User Boost

Users below `PhoenixRankerNewUserHistoryThreshold` route to a different inference cluster. New users also get `NEW_USER_OON_WEIGHT_FACTOR` for out-of-network content if they follow enough accounts.

Engagement History Window

128 engagement actions (`history_seq_len=128`), up to 32 candidates at once (`candidate_seq_len=32`). RoPE positional encoding with right-anchored option giving newest history items fixed positions.

9. Final Playbook

TOP DO's:

1. Trigger multiple engagement types (likes + replies + reposts + shares)
2. Post qualifying-length video for the exclusive VQV scoring bonus
3. Maximize dwell time with substantive, detailed content
4. Drive DM shares and copy-link shares (3 separate scoring signals)
5. Optimize for the follow action -- $P(\text{follow_author})$ is in the formula
6. Post consistently to build engagement history patterns
7. Stay in your niche for retrieval model clarity

TOP DON'Ts:

1. Trigger blocks/mutes/reports (strongest negative weights)
2. Get flagged by Grox safety classifiers (content removed entirely)
3. Post low-quality content (quality < 0.4 flagged, slop detection is real)
4. Flood with posts (author diversity decay is exponential)
5. Create scroll-past content (`not_dwelled` trains the model against you)